# DESCENT OF REGULARITY ALONG FPPF MORPHISMS

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites one same-chapter descent lemma, one Properties of Schemes lemma and one Algebra lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

*Allowed standard prerequisites (editorial).*

- fpqc descent of local Noetherianity. Stacks Project, Descent on Spaces, Lemma lemma-descend-locally-Noetherian. Shows $Y$ is locally Noetherian.
- Characterization of regular schemes. Stacks Project, Properties of Schemes, Lemma properties-lemma-characterize-regular. Reduces regularity of $Y$ to regularity of the local rings of the chart $V$.
- Flat morphisms under regular schemes. Stacks Project, Algebra, Lemma algebra-lemma-flat-under-regular. Concludes the local rings of $V$ are regular from those of $U$.

*Source reference.* Original lemma 06MJ: sources/stacks-chow/tags.txt, source lines 8503–8503. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02IT: sources/stacks-chow/tags.txt, source lines 3312–3312. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00OF: sources/stacks-chow/tags.txt, source lines 893–893. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $f : X \to Y$ be a morphism of algebraic spaces. If $f$ is locally of finite presentation, flat, and surjective and $X$ is regular, then $Y$ is regular.*

*Proof.* By Lemma 06MJ we know that $Y$ is locally Noetherian. Choose a scheme $V$ and a surjective étale morphism $V \to Y$. It suffices to prove that the local rings of $V$ are all regular local rings, see Properties, Lemma 02IT. Choose a scheme $U$ and a surjective étale morphism $U \to X \times_Y V$. As $f$ is surjective and flat the morphism of schemes $U \to V$ is surjective and flat. By assumption $U$ is a regular scheme in particular all of its local rings are regular (by the lemma above). Hence the lemma follows from Algebra, Lemma 00OF. □

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